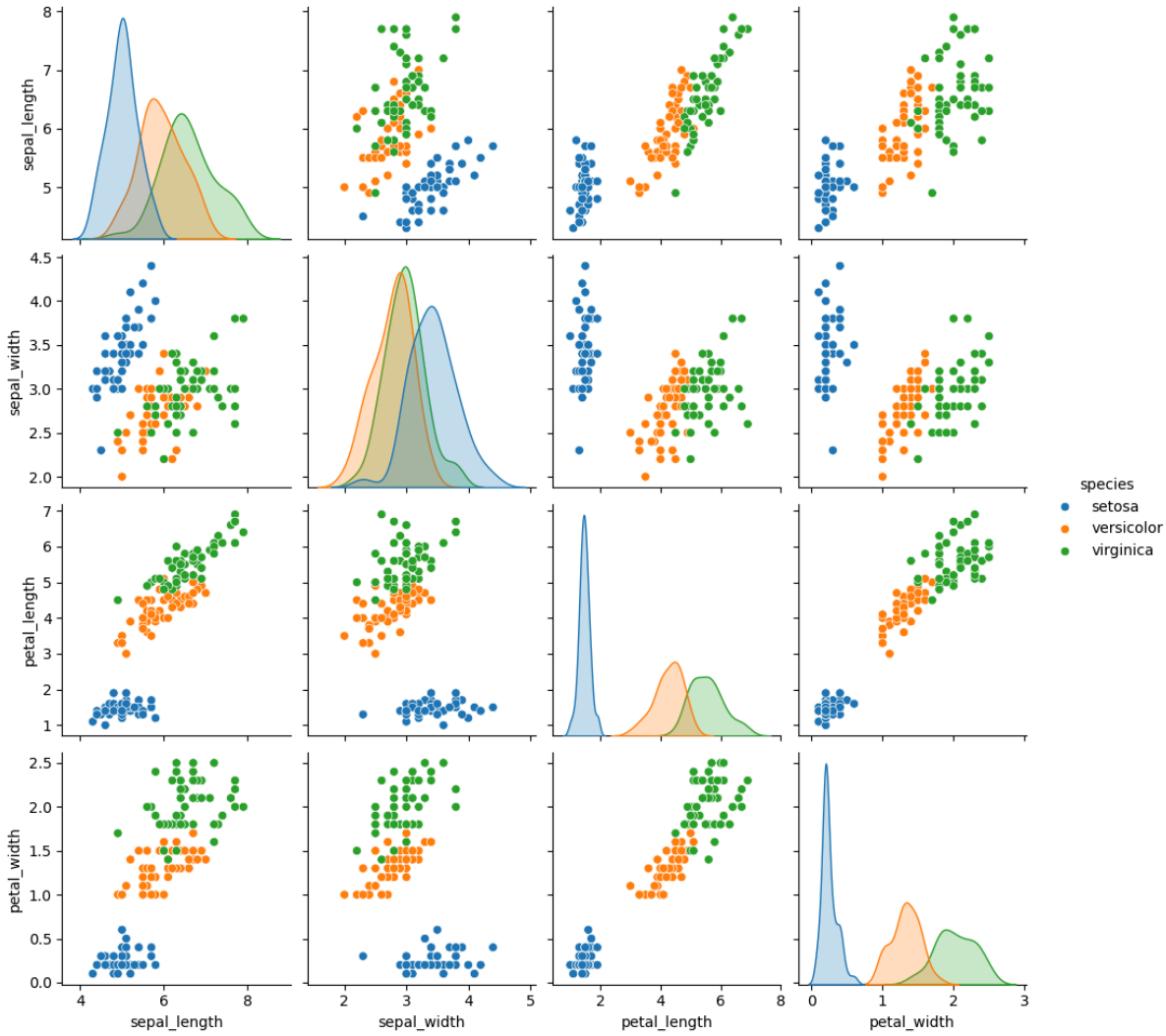
```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score, classification_report, confusion_matri
```

```
# 1. Load dataset directly from your screenshot link - No download needed
url = "https://raw.githubusercontent.com/mwaskom/seaborn-data/master/iris.csv"
df = pd.read_csv(url)
print(df.head())
```

```
   sepal_length  sepal_width  petal_length  petal_width species
0           5.1          3.5           1.4          0.2  setosa
1           4.9          3.0           1.4          0.2  setosa
2           4.7          3.2           1.3          0.2  setosa
3           4.6          3.1           1.5          0.2  setosa
4           5.0          3.6           1.4          0.2  setosa
```

```
# 2. EDA - Pairplot they asked for
sns.pairplot(df, hue='species')
plt.show()
```




```
# 3. Preprocessing - As per your task screenshot
X = df[['sepal_length','sepal_width','petal_length','petal_width']]
y = df['species']
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

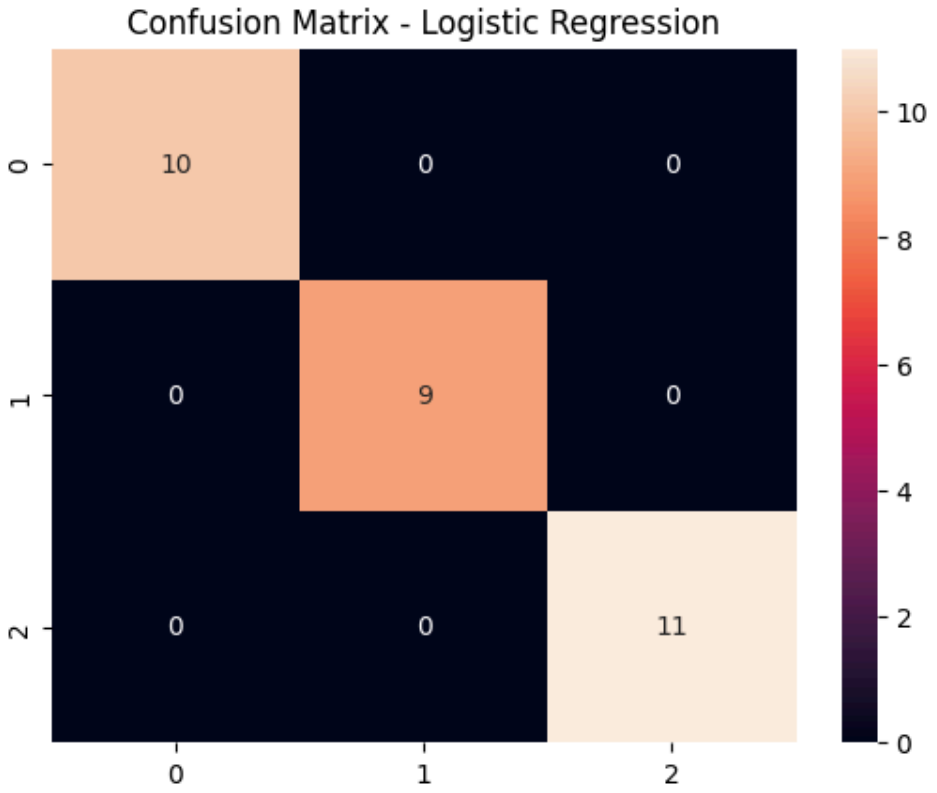
```
# 4. Split 80/20 as per task
X_train, X_test, y_train, y_test = train_test_split(X_scaled, y, test_size=0.2, ra
```

```
# 5. Train 2 models as per task
model1 = LogisticRegression()
model1.fit(X_train, y_train)
y_pred1 = model1.predict(X_test)
print("Logistic Regression Accuracy:", accuracy_score(y_test, y_pred1))
print(classification_report(y_test, y_pred1))
sns.heatmap(confusion_matrix(y_test, y_pred1), annot=True, fmt='d')
plt.title("Confusion Matrix - Logistic Regression")
plt.show()
```

```
Logistic Regression Accuracy: 1.0
              precision    recall  f1-score   support

      setosa       1.00      1.00      1.00        10
  versicolor       1.00      1.00      1.00         9
   virginica       1.00      1.00      1.00        11

    accuracy                           1.00        30
   macro avg       1.00      1.00      1.00        30
weighted avg       1.00      1.00      1.00        30
```




```
model2 = DecisionTreeClassifier()
model2.fit(X_train, y_train)
y_pred2 = model2.predict(X_test)
print("Decision Tree Accuracy:", accuracy_score(y_test, y_pred2))
print(classification_report(y_test, y_pred2))
sns.heatmap(confusion_matrix(y_test, y_pred2), annot=True, fmt='d')
plt.title("Confusion Matrix - Decision Tree")
plt.show()
```

```
Decision Tree Accuracy: 1.0
              precision    recall  f1-score   support

      setosa       1.00      1.00      1.00        10
  versicolor       1.00      1.00      1.00         9
   virginica       1.00      1.00      1.00        11
```